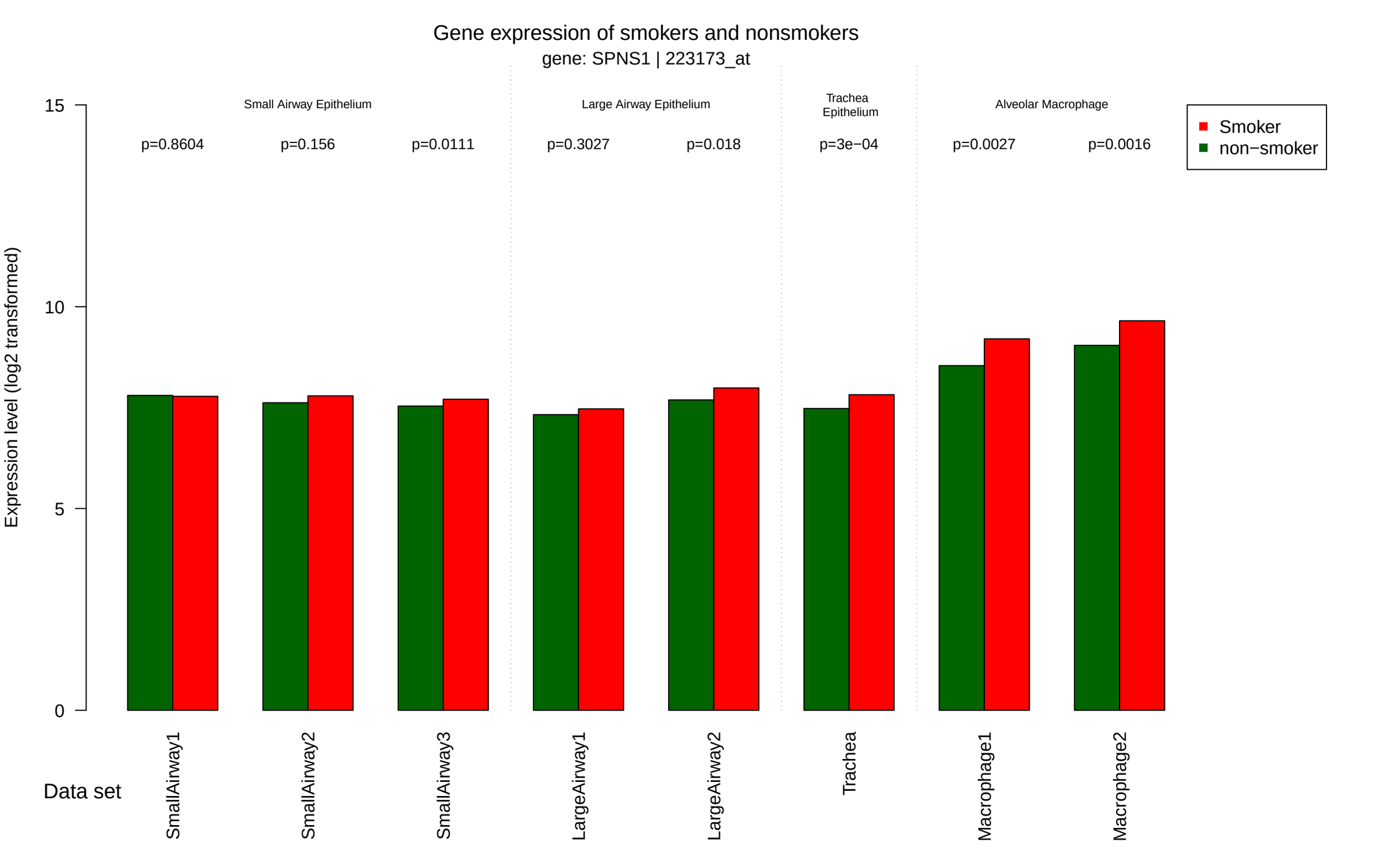
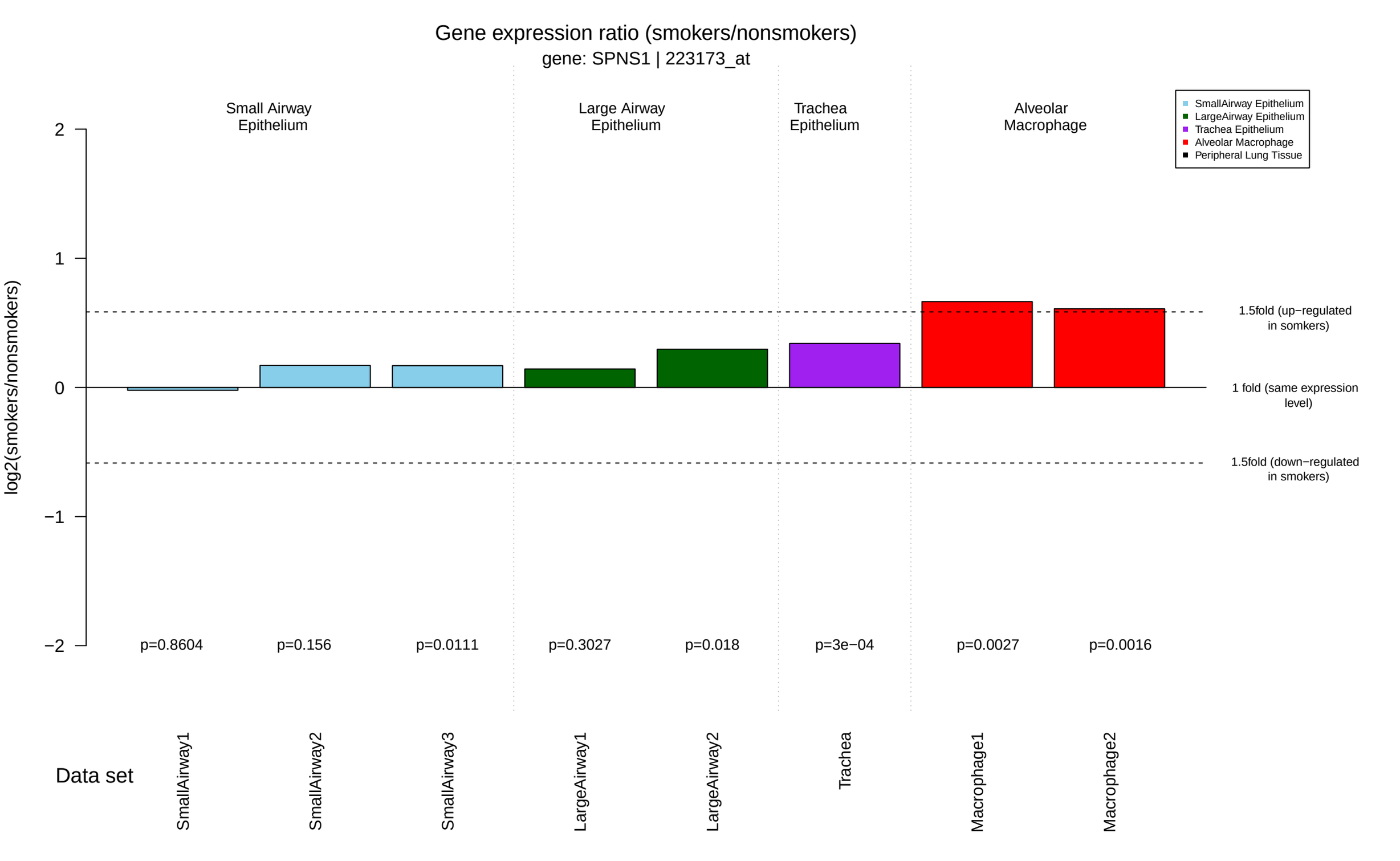
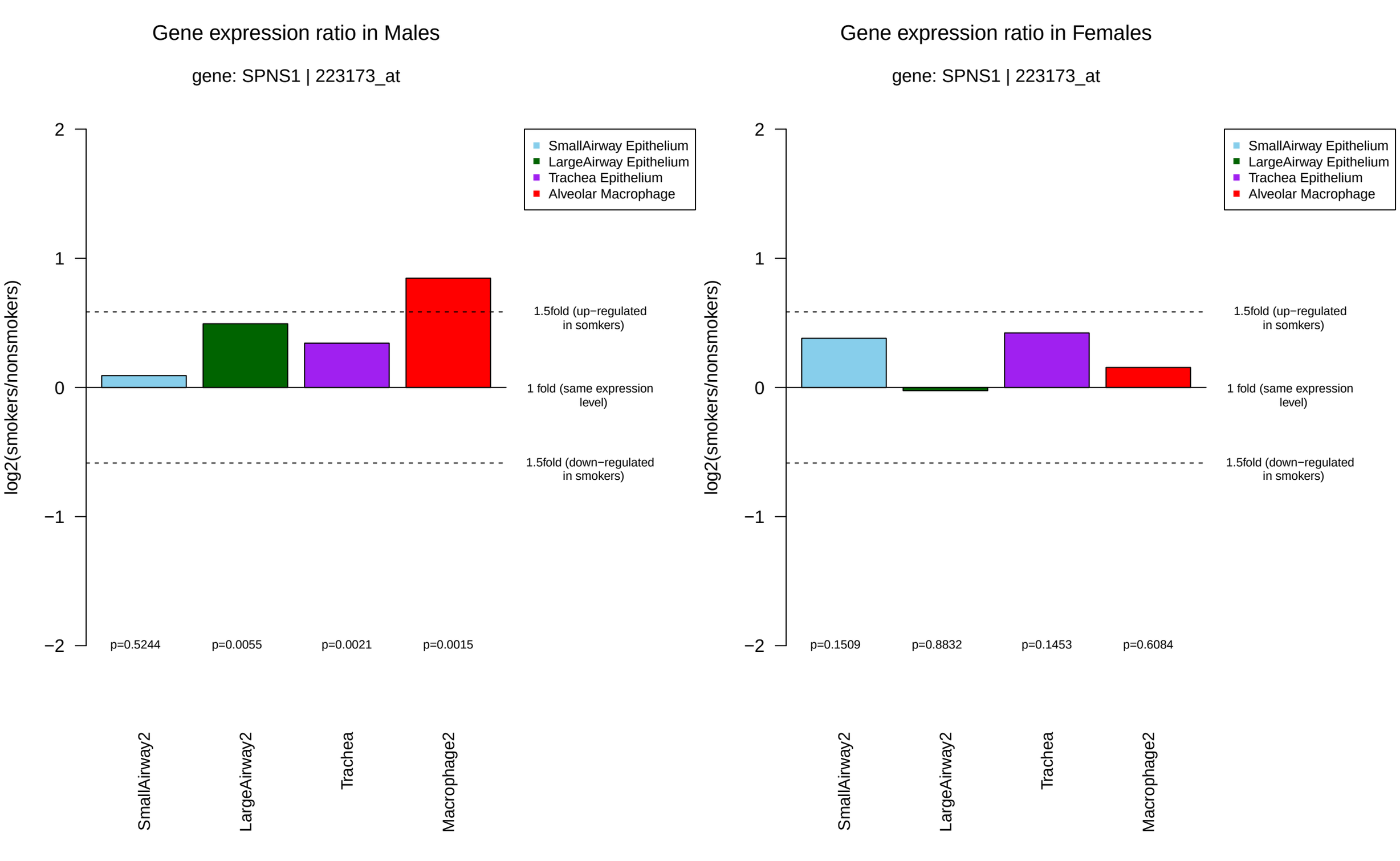
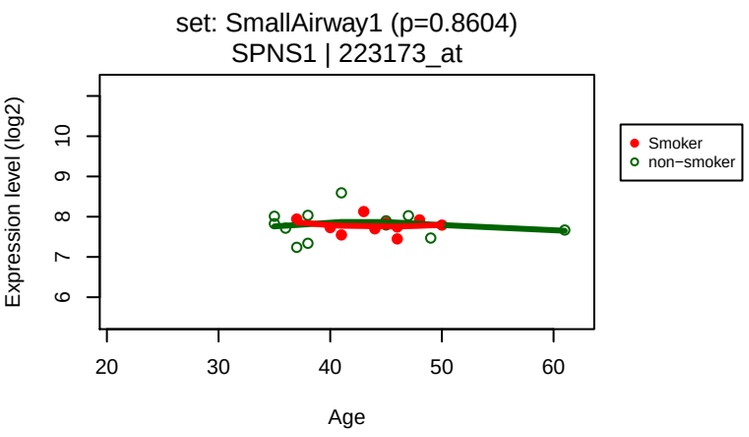
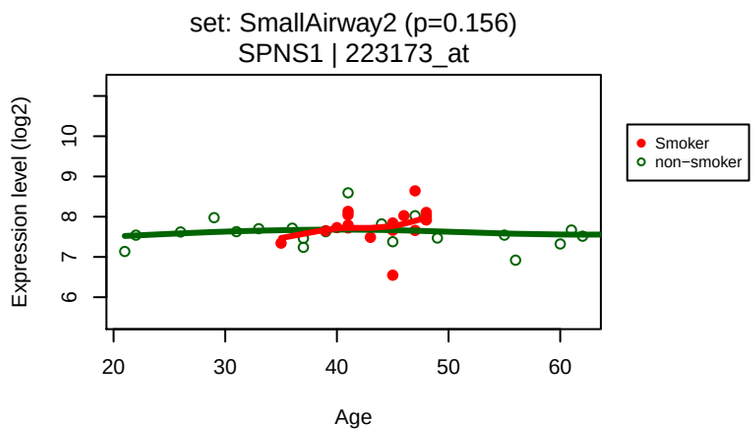
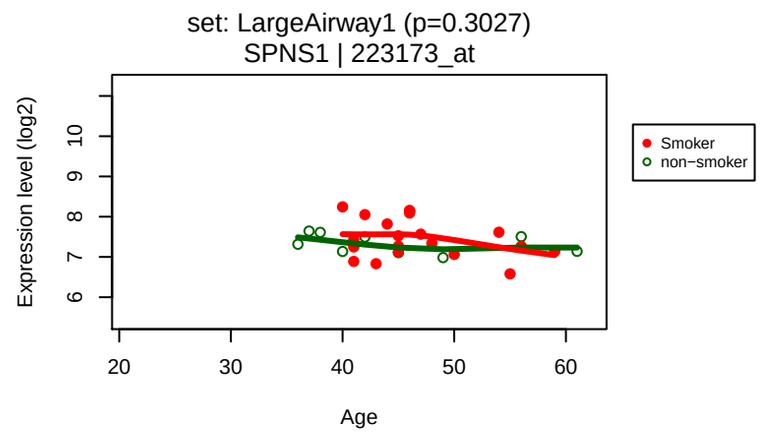

Gene expression of smokers and nonsmokers
gene: SPNS1 | 223173_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8604
p=0.156
p=0.0111
p=0.3027
p=0.018
p=3e−04
p=0.0027
p=0.0016
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

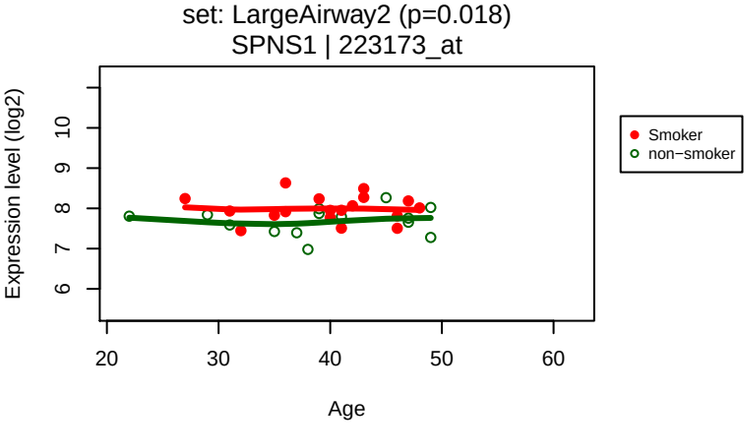

Gene expression ratio (smokers/nonsmokers)
gene: SPNS1 | 223173_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8604
p=0.156
p=0.0111
p=0.3027
p=0.018
p=3e−04
p=0.0027
p=0.0016
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

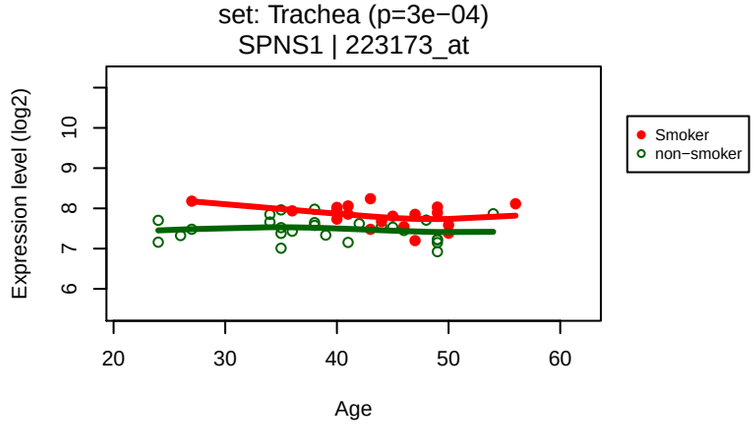

Gene expression ratio in Males
gene: SPNS1 | 223173_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5244
p=0.0055
p=0.0021
p=0.0015
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPNS1 | 223173_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1509
p=0.8832
p=0.1453
p=0.6084
SmallAirway2
LargeAirway2
Trachea
Macrophage2

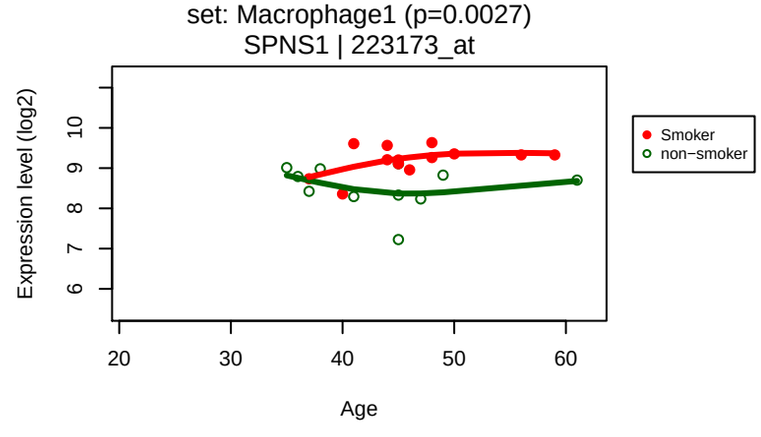

set: SmallAirway1 (p=0.8604)
SPNS1 | 223173_at
Expression level (log2)
Age
Smoker
non−smoker

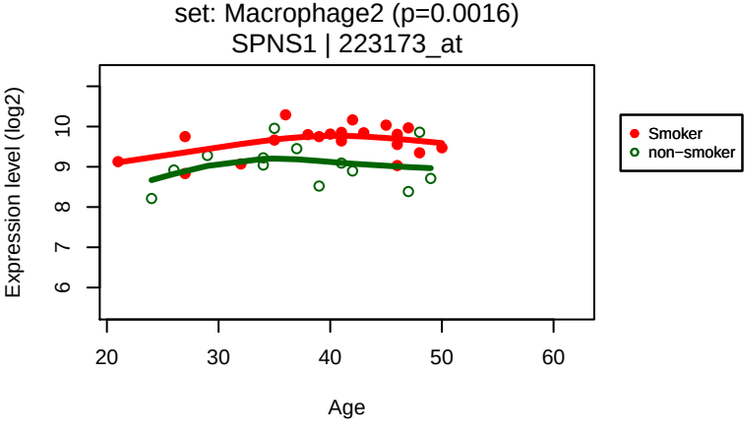

set: SmallAirway2 (p=0.156)
SPNS1 | 223173_at
Expression level (log2)
Age
Smoker
non−smoker

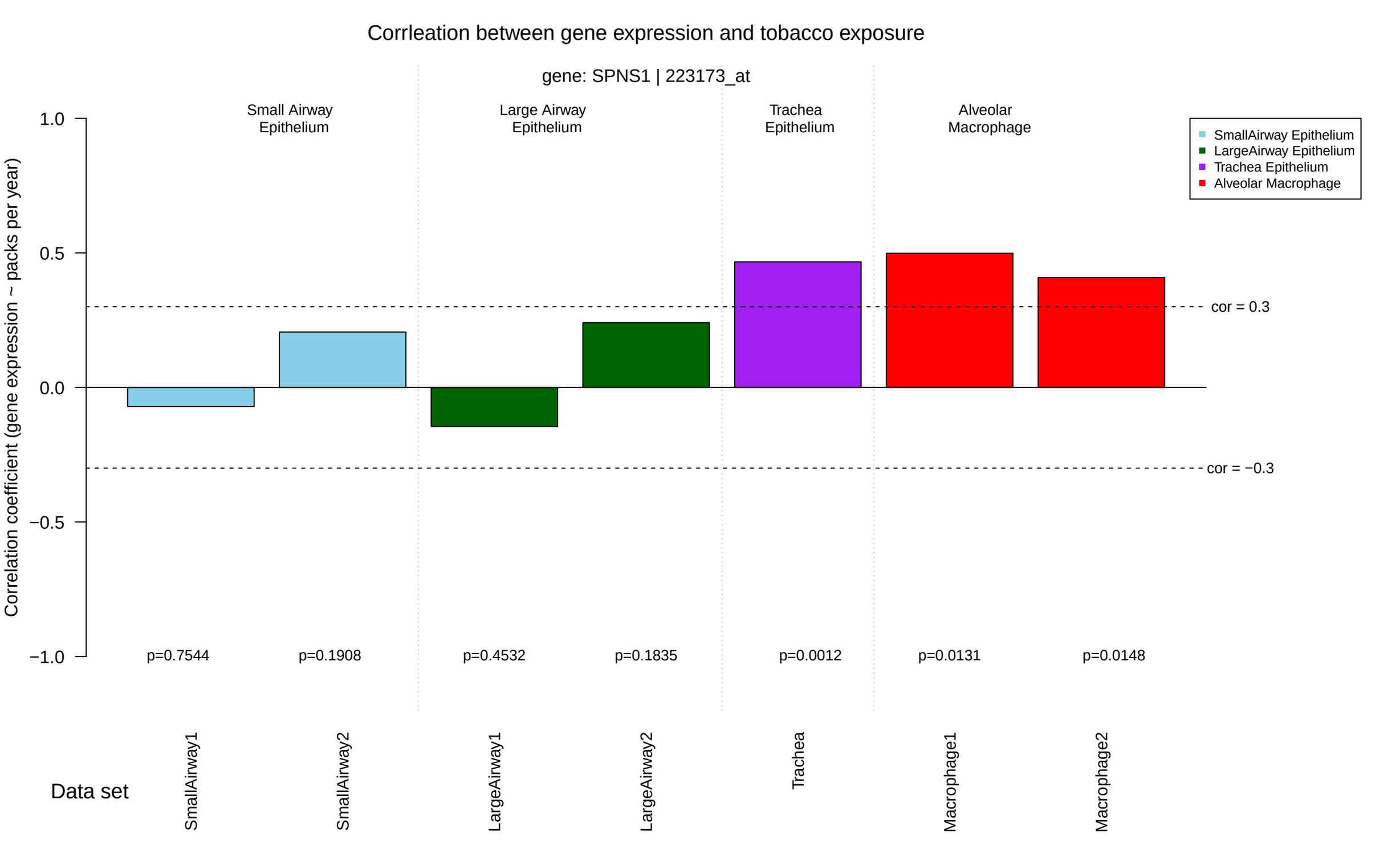

set: LargeAirway1 (p=0.3027)
SPNS1 | 223173_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.018)
SPNS1 | 223173_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=3e−04)
SPNS1 | 223173_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0027)
SPNS1 | 223173_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0016)
SPNS1 | 223173_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPNS1 | 223173_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7544
p=0.1908
p=0.4532
p=0.1835
p=0.0012
p=0.0131
p=0.0148
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2